MERMAID GeoCSV

SCHEMA COMPARISON / AUTOMAID v4.5.0 TO v4.10.0

Software versions are independent of the current payload schema, **1.0.0-rc1**.

Item	v4.5.0	v4.10.0
Container/header	`#dataset: GeoCSV`; no schema declaration	`# dataset: GeoCSV 2.0`; `# schema: MERMAID GeoCSV v1.0.0-rc1`
Project metadata	Attribution, MATLAB reader, pressure/depth and response comments	Producer, publisher, hoster and consumer descriptions
Columns	16 columns, including `StartTime`, `TimeDelay`, `TimeCorrection`	Same count/order; renamed to `Time`, `ClockOffset`, `TimeCorrectionApplied`
Position identifier	`Measurement:GPS:<sensor>`	`Measurement:Position:<sensor>`
Pressure identifier	`Measurement:Pressure:<sensor>`	`Measurement:Pressure-Absolute:<sensor>`
Recording identifier	`Algorithm(event):automaid:<version>`	`Derivation:Recording:automaid-<version>`
Crossing identifiers	`Algorithm(thermocline):automaid:<version>` for both directions	`Interpolation:Crossing-Descent:automaid-<version>` and `Interpolation:Crossing-Ascent:automaid-<version>`
Missing values / blank Location	`nan` placeholders; blank waveform Location emitted empty	Empty cells for absence; known blank Recording Location is `" "` (two spaces)
Instrument / elevation	`MERMAIDHydrophone(<model>)`; elevation absent	`MERMAID_<model>`; Recording elevation explicitly `0` metres (sea-level reference)
Time / numeric formatting	Millisecond timestamps; float32 conversion; SampleRate to one decimal	Six-digit UTC timestamps, ties-to-even microsecond rounding; float64 formatting; SampleRate to six decimals
Clock meaning	GPS delay and already-applied Recording correction	Same source signs/meaning: offset is instrument minus GPS time; correction is already included in Recording `Time`
Row selection	Only pressure rows explicitly restricted to the open first/last GPS interval	All rows within deployment `[begin, end)` and inclusive retained GPS span; Init/IcCycle measurements retained, derived rows excluded
Audit outputs	No clash reports or excluded-row sidecar	Raw-event clash reports, SHA-256-bound `geocsv-clash.json`, and local `outside_filterDate_gps.geocsv`

Sources: scripts/geocsv.py at tags v4.5.0 and v4.10.0; mermaid-geocsv/docs/format-spec.md; automaid/docs/geocsv.md and docs/clock-timing-matrix.md. Current consumers reject legacy names and missing-value conventions; renaming columns alone is insufficient migration.

AI-written documentation - non-authoritative. Verify against the source and schema before relying on it.